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Review Article

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Efficacy and safety of genicular nerve ablation techniques for knee osteoarthritis: a systematic review and meta-analysis of sham-controlled randomized trials

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Abstract

Objective: To evaluate the efficacy and safety of genicular neurolysis techniques, including radiofrequency ablation (RFA) and cryoneurolysis, versus sham procedures in adults with knee osteoarthritis (OA) ineligible for arthroplasty.

Methods: This systematic review and meta-analysis included randomized controlled trials (RCTs) comparing cooled, monopolar, or bipolar RFA, or cryoneurolysis, to sham treatment in adults with symptomatic knee OA. Studies with pulsed RFA, active comparators, or lacking pain or function outcomes were excluded. PubMed, Embase, and Cochrane CENTRAL were searched from inception to April 15, 2025, without language restriction. Risk of bias was assessed using the Cochrane RoB 2 tool. Random-effects meta-analysis calculated mean differences (MD) with 95% confidence intervals (CI). Heterogeneity was quantified with I^2, and certainty of evidence was rated using GRADE. Trial Sequential Analysis (TSA) assessed conclusiveness.

Results: Eight RCTs ($n = 627$) met eligibility criteria. Compared with sham, ablative techniques significantly reduced pain at 12 weeks (MD −1.65; 95% CI, −2.57 to −0.74; $I^2 = 83\%$) and improved WOMAC function scores (MD −11.37; 95% CI, −18.83 to −3.91; $I^2 = 91\%$). Subgroup analysis favored conventional monopolar RFA over other modalities. TSA confirmed conclusive benefits for pain at 12 weeks and function at 24 weeks. No serious adverse events were reported. Limitations include high heterogeneity, variable protocols, and limited long-term data.

Conclusions: Genicular neurolysis, particularly conventional monopolar RFA, provides clinically meaningful pain and functional improvement in knee OA with a favorable safety profile. Standardization of techniques and further trials are needed to confirm long-term outcomes.

Registration: PROSPERO CRD420251044248

Keywords: knee osteoarthritis; radiofrequency ablation; cryoneurolysis; chronic pain; systematic review; meta-analysis.

Introduction

Knee osteoarthritis (OA) is a leading cause of chronic pain and disability worldwide, particularly affecting older adults, and significantly contributes to the healthcare burden and reduced quality of life.[1] Conservative therapies, such as pharmacological agents and physical rehabilitation, remain the first-line treatments for managing knee OA. However, they often fail to achieve durable analgesia, especially in patients ineligible for arthroplasty.[2]

Conventional monopolar radiofrequency ablation (CRFA) of the genicular nerves has emerged as a minimally invasive alternative for pain control in patients with knee OA. Variants such as cooled RFA (CRFA), bipolar RFA, and cryoneurolysis differ in lesion geometry and biophysical mechanisms. CRFA produces a forward-projecting lesion shape due to internal cooling, which may enhance target coverage in certain anatomical approaches,[3] but it does not inherently produce a larger lesion volume than conventional monopolar probes of larger gauge or bipolar configurations. Bipolar RFA can generate substantially wider lesions, and larger-gauge conventional monopolar electrodes (eg, 16G) can create larger volumes than CRFA at lower cost. The purported advantage of CRFA is thus primarily geometric rather than volumetric.[4] Previous meta-analyses, including a network analysis by Wu et al., aggregated ablative and non-ablative techniques, such as pulsed RFA, intra-articular applications, and nerve blocks, into a single analytical framework.[5] However, these modalities differ fundamentally in mechanisms, lesion permanence, and clinical utility.

We hypothesized that ablative neurolytic interventions, including CRFA, bipolar RFA, conventional monopolar RFA, and cryoneurolysis, would provide sustained outcomes and should be analyzed independently. In this context, we conducted a systematic review and randomized controlled trials (RCTs) comparing ablative neurolytic interventions with sham procedures for the treatment of knee OA. We aimed to evaluate their efficacy in pain relief, functional improvement, patient satisfaction, and the durability of these outcomes.

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Methods

Study design

This systematic review and meta-analysis were conducted in accordance with the Cochrane Handbook for Systematic Reviews of Interventions[6] and reported in line with the PRISMA 2020 guidelines.[7] The review protocol was prospectively registered in the PROSPERO database (CRD420251044248).

Study eligibility

We included RCTs enrolling adults with symptomatic knee OA, diagnosed clinically and/or radiographically, irrespective of disease duration, severity, or setting. We excluded patients with a history of total or partial knee arthroplasty, or those currently considered candidates for arthroplasty. No exclusions were applied based on the type of image guidance or its absence. Eligible interventions comprised genicular neurolysis using conventional monopolar RFA, CRFA, bipolar RFA, or cryoneurolysis. Trials were required to compare these interventions with a sham (placebo) procedure without active neurolysis and to report at least one patient-centered outcome, such as pain intensity (numeric rating scale, verbal numeric rating scale, or visual analog scale) or physical function (eg, WOMAC function subscale). Trials using chemical neurolysis (phenol, alcohol) were excluded because lesion reproducibility is less consistent than with thermal or cryogenic ablation, given the variable dispersion of solutions and heterogeneous neuroanatomy. We excluded studies with non-randomized or observational designs; those using co-interventions in the control group (eg, intra-articular injections); protocol-only publications; trials assessing pulsed radiofrequency, and studies lacking pain or function as predefined outcomes. Additionally, trials directly comparing ablative methods with pharmacologic nerve blocks were excluded. Trials were considered eligible regardless of whether a diagnostic genicular nerve block was required as an entry criterion.

Search strategy

We systematically searched PubMed, Embase, and CENTRAL from inception to April 15, 2025, with the following search terms: ("radiofrequency ablation"[MeSH Terms] OR "radiofrequency ablation"[All Fields] OR "cooled radiofrequency"[All Fields] OR "bipolar radiofrequency"[All Fields] OR "cryoneurolysis"[All Fields] OR "cryoneuroablation"[All Fields]) AND ("knee pain"[MeSH Terms] OR "knee osteoarthritis"[MeSH Terms] OR "knee osteoarthritis"[All Fields]) AND (randomized controlled trial[pt] OR randomized[tiab] OR placebo[tiab] OR trial[tiab] OR randomly[tiab]). We manually screened the reference lists of the included articles and relevant reviews to identify additional eligible studies. No language restrictions were applied. Two authors (R.B. and A.J.B.) independently extracted data using predefined criteria for search, data extraction, and quality assessment. Disagreements between the authors were resolved by consensus. If consensus could not be reached, a third reviewer (B.Q.), with expertise in statistical analysis and methodological rigor, was designated to adjudicate disagreements. The entire screening process is documented in a PRISMA 2020 flow diagram, including detailed reasons for full-text exclusions.

Data extraction

Two reviewers (B.Q. and D.M.) independently extracted data from each included study using a standardized spreadsheet developed for this review. The extracted study characteristics included author, year, sample size, intervention details, patient demographics, outcome measures, follow-up time points, and summary statistics such as mean, standard deviation, and sample size.

For studies that reported unique outcome instruments not used in other trials, such as the Oxford Knee Score (OKS) or the knee injury and Osteoarthritis Outcome Score (KOOS), specific data were excluded from the quantitative synthesis to ensure consistency across pooled analyses. However, shared outcomes, such as pain, function, and patient satisfaction, from these studies were retained in the meta-analysis.

For each included trial, we also recorded procedural characteristics, including the type of image guidance used. All studies employed image guidance to target the genicular nerves, using either fluoroscopy, ultrasound, or both, as reported by the original authors.

Risk of bias assessment

The risk of bias for each included RCT was independently assessed using the Cochrane Risk of Bias 2.0 (RoB 2) tool.[8] The assessments were performed by a senior biostatistician in collaboration with one of the authors (R.B.). Each domain of the randomization process, deviations from intended interventions, missing outcome data, measurement of the outcome, and selection of the reported results were rated as having a low risk of bias, some concerns, or high risk of bias. For patient-reported outcomes (eg, pain measured by VAS or NRS), the patient was considered the outcome assessor. In accordance with Cochrane RoB 2 guidance, issues related to lack of participant blinding were assessed under "deviations from intended interventions" rather than "measurement of the outcome."

We extracted technical details from each study, including electrode size, number of nerves targeted, and number of lesions per nerve. Subgroup analysis was prespecified for electrode size, as the number of targeted nerves (typically 3) and the number of lesions per nerve (3) were largely consistent across studies.

Data synthesis

Data items and outcomes

The primary outcome of interest was pain intensity, measured using a numeric rating scale (NRS), verbal numeric rating scale (VNRS), or visual analog scale (VAS) at 2, 4, 12, 24, and 52 weeks postintervention. The secondary outcome was physical function, assessed primarily using the WOMAC function subscale.[9] When reported, treatment satisfaction was extracted as a tertiary outcome through the Global Perceived Effect Scale (GPES).[10]

Outcome data were abstracted at the defined follow-up points. When standard deviations (SDs) were not reported in the original studies, they were estimated using established statistical formulas. Specifically, when standard errors (SEs) were available, SDs were calculated using the formula: $SD = SE \times \sqrt{n}$, where n is the sample size of the group. When 95% confidence intervals (CI) were reported instead, SDs were derived using the formula: $SD = (\text{Upper Limit} - \text{Lower Limit})/(2 \times t)$, where t is the critical value from the t-distribution corresponding to the desired confidence level and degrees of freedom ($df = n - 1$). For larger samples ($n > 60$), the normal approximation $t \approx 1.96$ was applied.

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These conversions ensured standardized variance estimates across studies for meta-analysis pooling.

Effect measures

For continuous outcomes, MDs and their corresponding 95% CI were calculated to estimate the treatment effects between the intervention and sham groups. MDs were preferred whenever studies used the same scale (eg, NRS for pain and WOMAC for function). In cases where outcomes were reported using different instruments and pooling was still considered clinically meaningful, standardized MDs (SMDs) were explored in sensitivity analyses.

When studies reported changes from baseline scores without final values, they were included in the analysis, assuming comparable baseline characteristics between the groups. When necessary, SDs were calculated or imputed using available statistical parameters such as CI, *P* values, or interquartile ranges.

Certainty of evidence

The certainty of evidence for each key outcome (pain reduction, functional improvement, and patient satisfaction) was assessed by 2 reviewers (R.B.B. and A.J.B.) independently using the GRADE approach, evaluating risk of bias, inconsistency, indirectness, imprecision, and publication bias.[11,12] Disagreements were resolved through discussions.

The certainty of evidence was rated as high, moderate, low, or low based on the presence and severity of limitations in these domains. For imprecise outcomes, the width of the 95% CI and the optimal information size were considered. When trial sequential analysis (TSA) indicated that the required information size had not been reached, the certainty of the evidence was downgraded due to imprecision.

A summary of findings (SoF) table was generated to present the effect estimates, number of participants, heterogeneity measures (I^2), and GRADE rating for each main comparison and outcome.

Assessment of heterogeneity

Statistical heterogeneity across studies was assessed using the I^2 statistic, Cochran's Q test, and τ^2 (tau-squared) values. I^2 values above 50% were interpreted as substantial heterogeneity, and those exceeding 75% were considered indicative of considerable heterogeneity. When high heterogeneity was detected, potential sources were explored using subgroup and sensitivity analyses.

Visual inspection of the forest plots also supported the identification of heterogeneity patterns, and prediction intervals were generated to assess between-study variability and the range of expected effects in future trials.

Meta-analysis

All quantitative syntheses were performed using random-effects models, with the restricted maximum likelihood (REML) estimator for between-study variance (τ^2). The primary analysis focused on MDs with a 95% CI for each outcome and time point. Forest plots were generated to illustrate pooled effect sizes, individual study contributions, and overall heterogeneity.

Statistical analyses were performed using the *meta* and *metafor* packages in R software (version 4.3.1). Prediction intervals were included in forest plots to estimate the expected treatment effect in future trials. Study weighting was based on inverse variance, and effect sizes were calculated separately for each time point (2, 4, 12, and 24 weeks), allowing for time-specific interpretation.

Subgroup analyses

Subgroup analyses were conducted to explore whether the type of ablation technique influenced treatment effects. Meta-regression models were constructed using the type of intervention as a categorical moderator variable to compare conventional monopolar RFA, CRFA, bipolar RFA, and cryoneurolysis.

These analyses were performed separately for each outcome and time point, particularly at 12 and 24 weeks, using the *metafor* package in R. The heterogeneity explained by subgroup differences was quantified using the R^2 statistic. The subgroup results were interpreted in terms of consistency, statistical significance, and clinical plausibility.

TSA

TSA[13] was performed to determine whether the cumulative evidence was sufficient to confirm or reject the anticipated intervention effects, while accounting for repeated significance testing and potential random errors. The TSA was applied to the following main outcomes: pain intensity, functional improvement (WOMAC), and patient satisfaction (GPES) at various timepoints (2, 4, 12, and 24 weeks).

Analyses were conducted using the *rpact* package in R. Two-sided monitoring boundaries were applied, with a type I error (α) of 5% and a type II error (β) of 20% (power = 80%). For each outcome, the minimal clinically important difference (MCID) and pooled standard deviation were used to calculate the required information size (RIS). The cumulative Z-curve was compared with the sequential trial monitoring boundaries. TSA was used to evaluate whether the evidence was conclusive (crossing boundaries), inconclusive (not crossing boundaries), or suggestive of futility.

Results

Study selection

As illustrated in Figure 1, 4236 records were identified through database searches, and an additional 36 records were retrieved from other sources such as Google Scholar. After removing 30 duplicates, 4242 records were screened based on their titles and abstracts. Of these, 4200 were excluded because they did not meet the inclusion criteria, leaving 42 full-text articles assessed for eligibility. After full-text review, 34 studies were excluded. Ultimately, 8 RCTs[14–21] met the eligibility criteria and were included in the final meta-analysis.

Characteristics of the included studies

The 8 RCTs included in this review were published between 2011 and 2025, comprising a total of 627 patients with chronic knee pain due to OA. All studies compared genicular neurolysis procedures, including conventional monopolar RFA,[14–17] CRFA,[21] bipolar RFA,[19] and cryoneurolysis,[18,20] to a sham (placebo) control group. Across the 8 included trials, 7 employed image-guided techniques (fluoroscopy or ultrasound) to localize the superomedial (SMG), superolateral (SLG), and inferomedial (IMG) genicular nerves at standard periosteal landmarks. Only 1 study (Radnovich et al[18]) relied exclusively on anatomical landmarks without imaging

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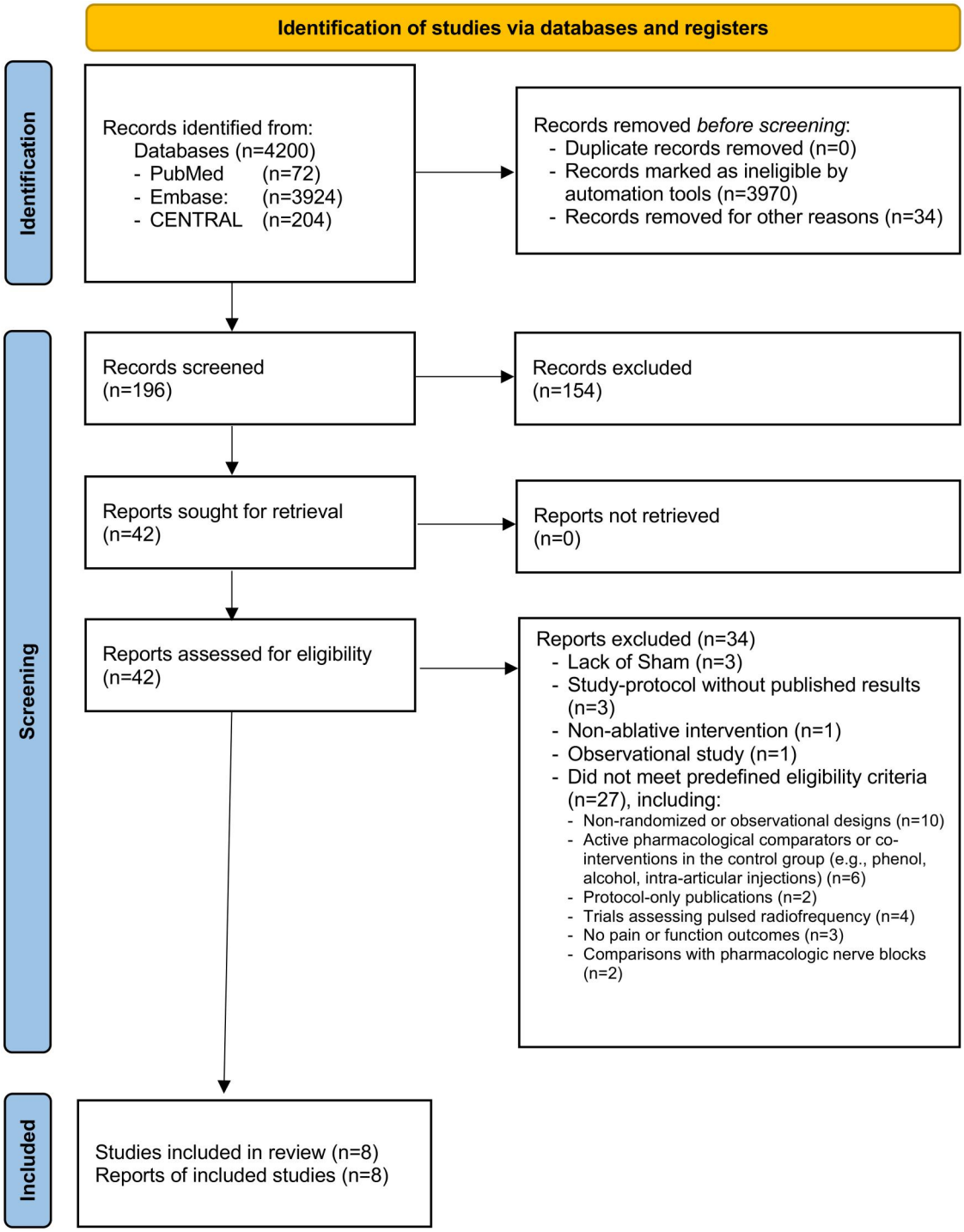

Figure 1. PRISMA flow diagram of study selection.

guidance to the treatment target, infrapatellar branch of the saphenous nerve (IPBSN).

The sample sizes ranged from 32 to 121. The mean age of participants was 64.1 years (58–70.5), and the mean body mass index was 28.3 (26–31.1) kg/m^2. The follow-up period varied from 2 weeks to 12 months, with most studies reporting outcomes at 2, 4, 12, or 24 weeks. In most trials, pain was assessed using numeric rating scales (NRS or VNRS), while function was evaluated using the WOMAC index. One study used the KOOS,[20] and another reported the OKS[14]; these studies were excluded from pain-related pooled analyses owing to incompatibility. Patient satisfaction data were reported in 4 trials.[15–17,21] Additional characteristics are presented in Table 1.

Risk of bias assessment

The risk of bias was assessed for each of the 8 included studies using the Cochrane Risk of Bias 2 (RoB 2) tool. Overall, 2 trials were judged to have low risk of bias across all 5 domains.[16,19] Five studies raised some concerns, primarily owing to unclear blinding procedures for the outcome assessors or incomplete reporting of the allocation methods.[14,17,18,20,21] One study was

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Table 1. Characteristics of included randomized controlled trials.

Study	No. of patients	Population	Intervention	Electrode size	Guidance method	Nerves target	Follow-up	Baseline age (mean) Int/Control	Female (%) Int/Control	Adverse events
Choi et al[14]	35	KL II-IV	RFA	22G	Fluoro	SMG, IMG, SLG	12 weeks	67.9/66.5	88/83	Minimal
Radnovich et al[18]	180	KL II-III	Cryoneurolysis	27G	Anatomical	IPBSN	24 weeks	60.6/61.3	65.3/67.8	Mild
El-Hakeim et al[17]	60	KL III-IV	RFA	22G	Fluoro	SMG, IMG, SLG	52 weeks	62/56.9	70/60	Mild
Malaithong et al[19]	60	KL III-IV	B-RFA	18G	Fluoro	SMG, IMG, SLG	24 weeks	64.8/68.6	78.1/84.4	Minimal
Kwon et al[21]	35	KL II-IV	C-RFA	17G	Fluoro	SMG, IMG, SLG	24 weeks	69.1/71.9	70/80	Few
Ma et al[15]	78	KL III-IV	RFA	22G	Ultrasound	SMG, IMG, SLG	24 weeks	65.1/63.9	82/85.7	Mild
Makkar et al[16]	54	KL II-IV	RFA	22G	Fluoro	SMG, IMG, SLG	12 weeks	58.3/57.7	74/59	Minimal
Nygaard et al[20]	87	KL II-IV	Cryoneurolysis	17G	Ultrasound	SMG, IMG, SLG	52 weeks	68.6/62.2	50/48.7	Mild

Abbreviations: KL, Kellgren–Lawrence; RFA, radiofrequency ablation; B-RFA, bipolar radiofrequency ablation; C-RFA, cooled radiofrequency ablation; SMG, superomedial genicular nerve; IMG, inferomedial genicular nerve; SLG, superolateral genicular nerve; IPBSN, infrapatellar branch of the saphenous nerve.

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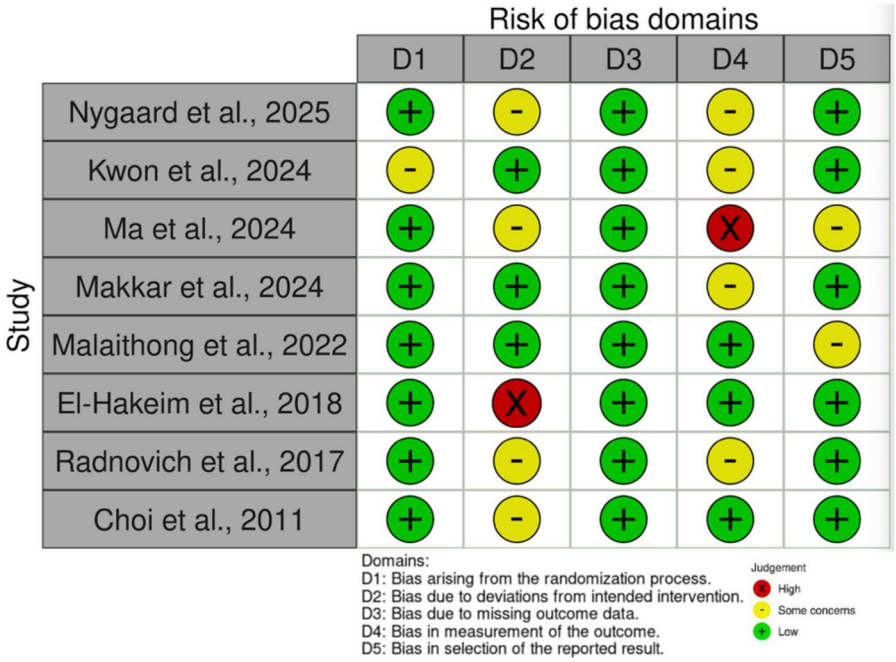

Figure 2. Risk of bias assessment using the Cochrane RoB 2 tool.

deemed to have a high risk of bias owing to insufficient blinding in the outcome assessment and uncertainty regarding selective reporting.[15]

Across studies, randomization processes were generally well described, and outcome data were largely complete, with low attrition rates. However, some trials lacked sufficient detail regarding the concealment of allocation and maintenance of blinding during outcome assessment,[15,16] particularly for subjective outcomes, such as pain and satisfaction. A visual representation of the risk of bias across domains and studies is presented in Figure 2.

Primary outcomes

Pain reduction

Pain scores were significantly lower in the intervention group at 2 weeks (MD = −1.15; 95% CI, −1.41 to −0.88; $P < .0001$; $I^2 = 29\%$), 4 weeks (MD = −1.80; 95% CI, −3.29 to −0.32; $P = .0174$; $I^2 = 85\%$), 12 weeks (MD = −1.65; 95% CI, −2.57 to −0.74; $P = .0004$; $I^2 = 83\%$), and 24 weeks (MD = −1.38; 95% CI, −2.47 to −0.29; $P = .0134$; $I^2 = 90\%$) than in the control group. The certainty of evidence for pain reduction at 12 weeks was rated as moderate, downgraded for inconsistency due to high heterogeneity ($I^2 = 83\%$). However, at 52 weeks, no significant difference was observed between groups (MD = 0.10; 95% CI, −3.91 to 4.11; $P = 0.961$; $I^2 = 0\%$) as shown in the forest plot (Figure 3).

Functional outcomes (WOMAC subscale)

Functional outcomes showed a statistically significant improvement in the intervention group at 2 weeks (MD = −9.57; 95% CI, −14.26 to −4.87; $P = .00006$; $I^2 = 0\%$). However, at 4 weeks, functional scores were comparable between groups (MD = −9.89; 95% CI, −21.75 to 1.98; $P = 0.104$; $I^2 = 93.5\%$). At 12 weeks, the intervention demonstrated a significant benefit (MD = −11.37; 95% CI, −18.83 to −3.91; $P = .0028$; $I^2 = 91.6\%$), which persisted at 24 weeks (MD = −9.25; 95% CI, −12.08 to −6.42; $P < .00001$; $I^2 = 57.5\%$). Functional improvement at 24 weeks was supported by moderate-certainty evidence, downgraded for imprecision given the wide CIs across trials (Figure 4).

Patient GPES

At 4 weeks, satisfaction scores were comparable between groups (MD = 0.85; 95% CI, −0.95 to 2.65; $P = 0.355$; $I^2 = 58.8\%$). Similarly, the results were not statistically significant at 12 weeks (MD = 0.63; 95% CI, −0.05 to 1.32; $P = .070$; $I^2 = 22.8\%$) and 24 weeks (MD = 0.68; 95% CI, −1.23 to 2.58; $P = 0.484$; $I^2 = 90.0\%$). Patient GPES were rated as low certainty, reflecting inconsistency and imprecision across the small number of available studies (Figure 5).

A summary of our findings is provided in Table S1.

Subgroup analysis by intervention technique

At 4 weeks, a subgroup analysis of 7 studies ($n = 478$) revealed a significant reduction in pain favoring neuroablative techniques over sham (MD = −1.80; 95% CI, −2.91 to −0.70; $I^2 = 84.6\%$). Among the techniques, conventional monopolar RFA demonstrated the most pronounced and consistent effect (MD = −2.69; 95% CI, −3.90 to −1.47; $P < .00001$; $I^2 = 87.1\%$), followed by cryoneurolysis (MD = −1.70; 95% CI, −2.59 to −0.81; $P = .00018$). Bipolar RFA and CRFA did not show significant benefits. The subgroup effect was statistically significant ($\chi^2 = 12.07$; df = 3; $P = .0072$), indicating differential efficacy between intervention modalities. The corresponding forest plots are shown in Figure S1.

At 12 weeks, pooled analysis demonstrated significant between-group differences ($P = .0048$). Conventional monopolar RFA remained the most effective technique (MD = −2.36; 95% CI, −3.07 to −1.64; $P < .00001$; $I^2 = 78.7\%$), whereas cryoneurolysis did not reach statistical significance (MD = −0.67; 95% CI, −1.45 to 0.11). Bipolar

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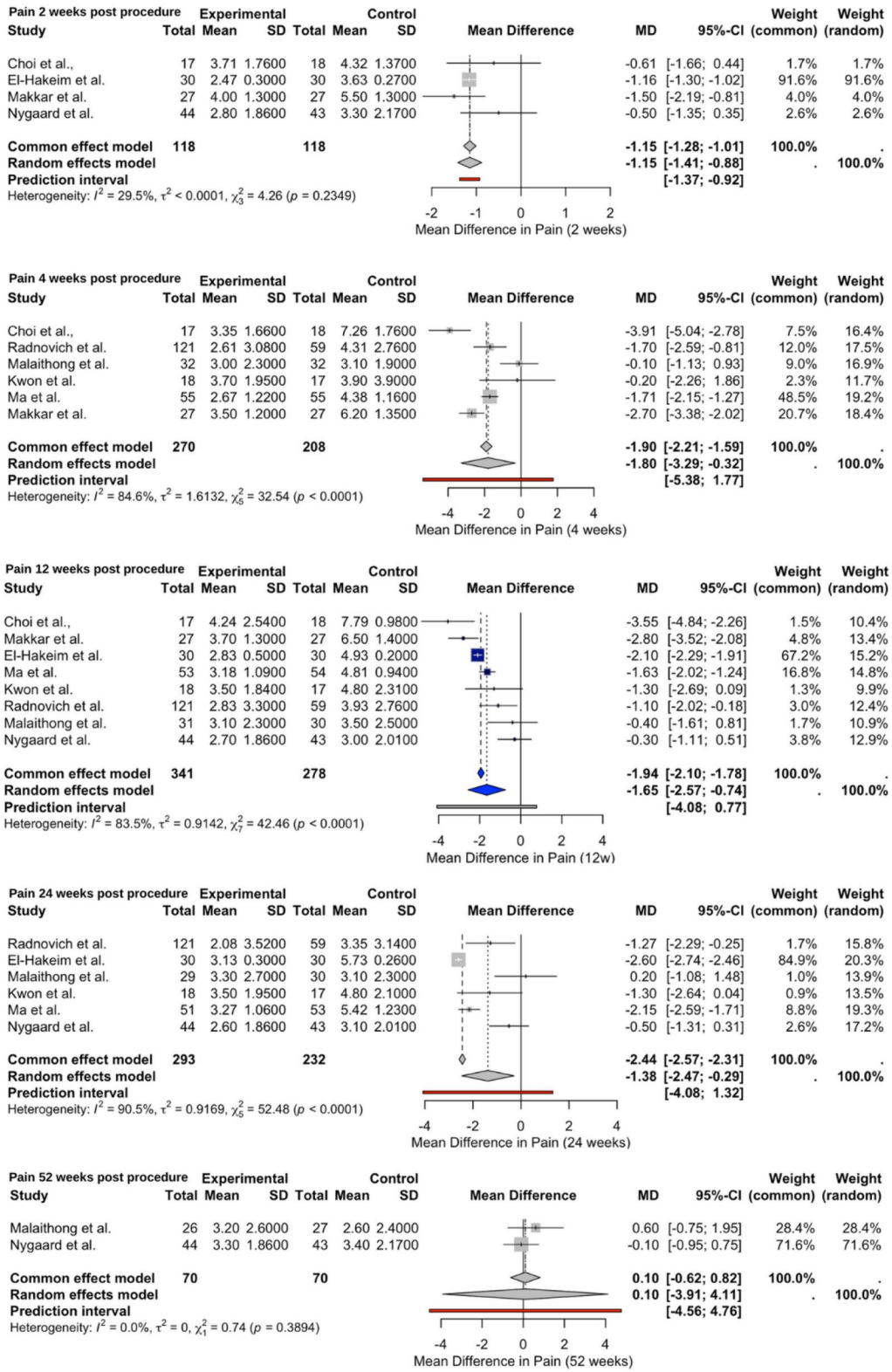

Figure 3. Significant pain reduction at 2, 4, 12, and 24 weeks for patients in the intervention group. No clinically meaningful difference is detected at 52 weeks.

RFA (MD = −0.40; 95% CI, −1.61 to 0.81) and CRFA (MD = −1.30; 95% CI, −2.69 to 0.09) were evaluated in single studies and yielded nonsignificant results (see Figure S2).

At 24 weeks, results continued to favor conventional monopolar RFA (MD = −2.43; 95% CI, −2.86 to −2.00; $P < .00001$), followed by cryoneurolysis (MD = −0.82; 95% CI, −1.57 to −0.08; $P = .032$). Bipolar RFA showed no meaningful effect (MD = 0.20; 95% CI, −1.08 to 1.48), and CRFA remained statistically nonsignificant (MD = −1.30; 95% CI, −2.64 to 0.04). Between-group differences were statistically

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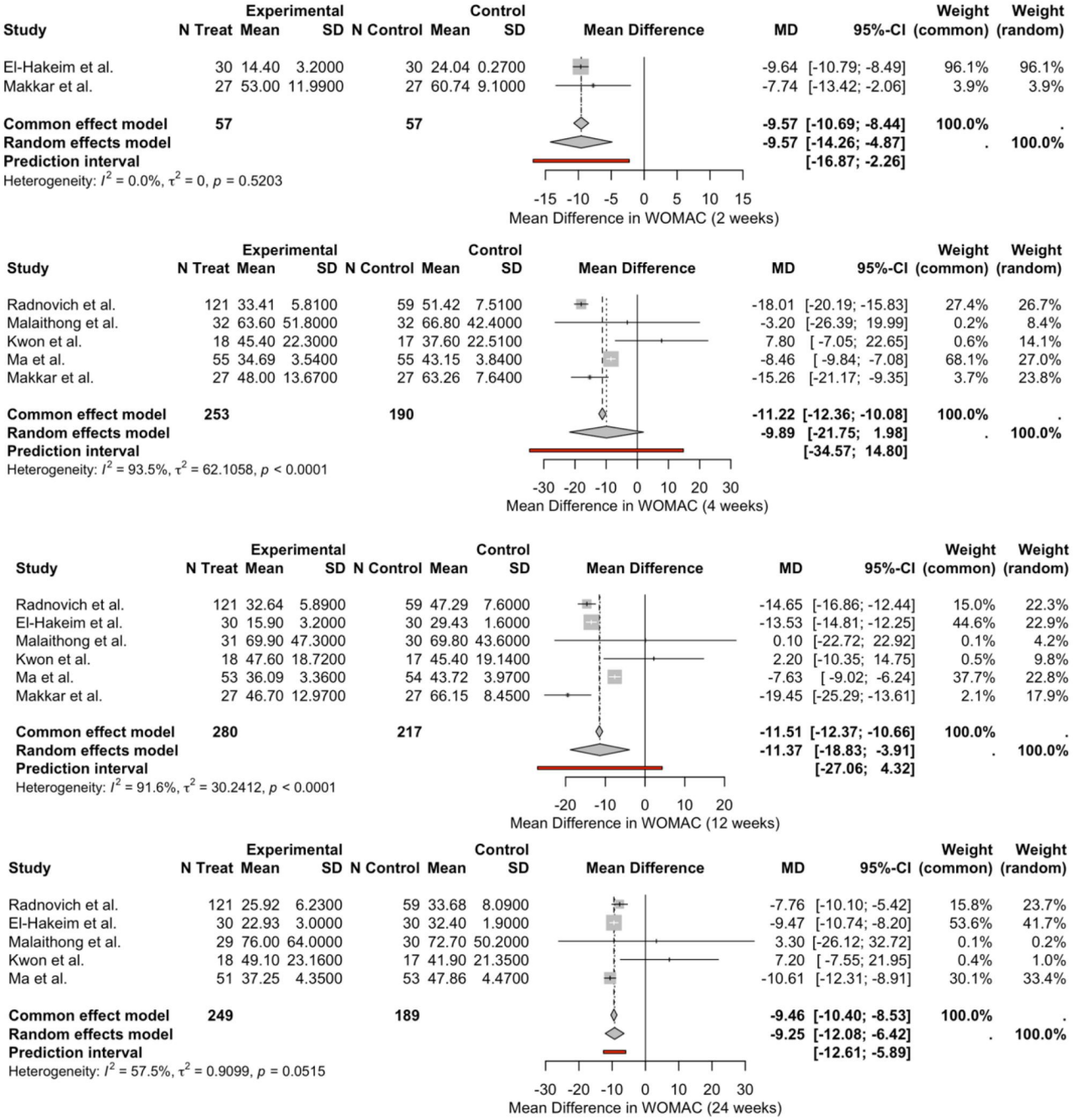

Figure 4. Statistically significant functional improvement for patients in the intervention group at 2, 12, and 24 weeks. There are no significant differences between groups at the 4-week time point.

significant (Q = 24.64; df = 3; $P < .0001$), reinforcing the technique-dependent effect on pain relief (Figure S3).

Although image guidance was used in nearly all included trials, Radnovich et al[18] relied exclusively on anatomical landmarks. Given that this was the only trial without image guidance, subgroup analysis was not feasible.

Functional outcomes by subgroup analysis

As shown in Figure S4, pooled data from 5 trials ($n = 443$) showed a significant improvement in function with neuroablative techniques versus sham at 4 weeks (MD = −9.89; 95% CI, −17.95 to −1.82; $P = .0162$), albeit with high heterogeneity (I^2 = 93.5%). Subgroup analysis identified conventional monopolar RFA as yielding the most consistent benefit (MD = −11.23; 95% CI, −17.78 to −4.68; $P < .00001$; I^2 = 79.3%). Other modalities (cryoablation, bipolar RFA, and CRFA) have been represented by single studies with heterogeneous results. A significant subgroup effect was observed ($P = .0012$).

At 12 weeks, 5 studies stratified by intervention technique again favored conventional monopolar RFA (MD = −13.07; 95% CI, −16.56 to −6.18; $P < .00001$; I^2 = 95.6%) and cryoneurolysis (MD = −14.65; 95% CI, −16.86 to −12.44; $P = .0001$). Bipolar RFA and CRFA showed no significant

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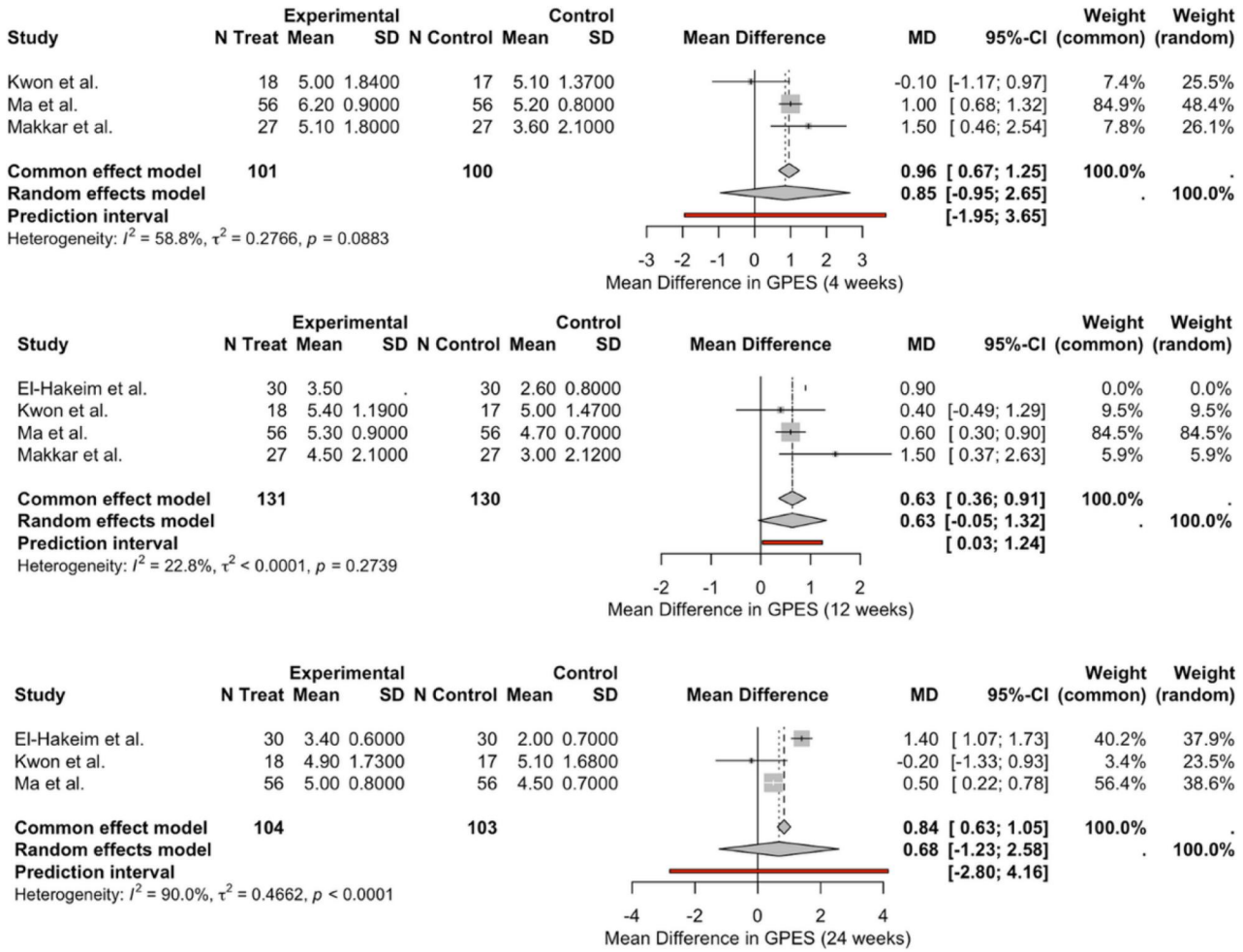

Figure 5. No significant patient satisfaction with treatment at 4, 12, and 24 weeks.

improvement. The differences between the groups remained significant ($P = .0407$), as illustrated in Figure S5.

Subgroup analysis of GPES at 4 weeks

At 4 weeks, pooled analysis of 3 RCTs indicated significantly greater patient-reported satisfaction with ablative interventions versus sham (MD = 0.85; 95% CI, 0.10–1.60; $P = .026$). Conventional monopolar RFA provided the most robust benefit (MD = 1.04; 95% CI, 0.74–1.34; $P < .000001$), while CRFA yielded no statistically significant effect (MD = −0.10; 95% CI, −1.17 to 0.97). The difference between the techniques was statistically significant ($P = .0443$), suggesting a higher perceived effectiveness of conventional monopolar RFA at early follow-up (Figure S6).

At 12 weeks, 4 studies were included. The overall effect remained statistically significant in favor of intervention (MD = 0.74; 95% CI, 0.20–1.28; $P = .0074$). Conventional monopolar RFA continued to show a consistent effect (MD = 0.79; 95% CI, 0.47–1.10; $P < .00001$), while CRFA did not reach statistical significance (MD = 0.40; 95% CI, −0.49 to 1.29). However, between-group differences were not significant ($P = 0.422$), indicating the absence of a subgroup effect at this time point (Figure S7).

Subgroup analysis by electrode size

Across the 8 included trials, 7 used the classical 3-nerve approach (superomedial, superolateral, and inferomedial), while 1 study targeted only 2 nerves.[18] Lesion number per nerve was uniform (1 lesion per nerve). Electrode sizes varied, with 2 studies using 17G,[20,21] 1 using 18G,[19] 4 using 22G,[14–17] and 1 using 27G electrodes.[18] Subgroup analyses stratified by electrode size did not demonstrate statistically significant differences in pain, function, or GPES outcomes, although studies employing 22G electrodes contributed to higher heterogeneity. Detailed results, including forest plots and subgroup statistics, are provided in Figures S8-S12.

TSA

To assess the robustness and conclusiveness of the cumulative evidence, we performed TSA for pain reduction and functional improvement at 12 and 24 weeks. A random-effects model was used, applying an O'Brien-Fleming α-spending function with a 2-sided 5% type I error and 80% statistical power.

For pain reduction at 12 weeks, the cumulative Z-curve crossed the conventional monopolar significance threshold and the O'Brien-Fleming monitoring boundary before the RIS of 148 participants was reached (Figure S13). The final Z-score remained substantially above the efficacy threshold,

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Table 2. Trial sequential analysis (TSA) of primary and secondary outcomes, showing whether monitoring boundaries were crossed and the corresponding interpretation.

Outcome—timepoint	TSA monitoring boundary crossed	TSA conclusion	Interpretation
Pain—02 weeks	NO	Inconclusive	More evidence required to confirm benefit.
Pain—04 weeks	NO	Inconclusive	Trend observed, but not statistically secured.
Pain—12 weeks	YES	Conclusive	Strong evidence for pain reduction.
Pain—24 weeks	NO	Inconclusive	Long-term effect not statistically confirmed.
WOMAC—12 weeks	NO	Inconclusive	Statistical significance not robust under TSA.
WOMAC—24 weeks	YES	Conclusive	Reliable evidence of functional improvement.
GPES - 12 weeks	NO	Inconclusive	Satisfaction trend not yet statistically verified.

Table 3. Ranking of interventions based on mean differences.

Intervention	Pain 02 weeks	Pain 12 weeks	Pain 24 weeks	Function 24 weeks	Rank pain 02 weeks	Rank pain 12 weeks	Rank pain 24 weeks	Rank function 24 weeks	Mean rank
Conventional RF	−1.09	−2.52	−2.38	−10.0	1	1	1	1	1.0
Cryoneurolysis	−0.5	−0.7	−0.885	−7.76	2	2	2	2	2.0
Cooled RF	–	−1.3	−1.3	7.2	–	4	3	3	3.33
Bipolar RF	–	−0.4	0.2	3.3	–	3	4	4	3.67

providing firm evidence to support the short-term analgesic benefits of ablative neurolysis.

For pain reduction at 24 weeks, although the overall effect estimate favored the intervention, the cumulative Z-curve did not cross the adjusted monitoring boundary (Figure S14). Moreover, the accrued sample size ($n = 334$) was below the RIS of 369 participants, suggesting that the long-term analgesic effect remains promising but inconclusive, warranting further studies.

Regarding functional improvement measured using the WOMAC, separate TSA evaluations were conducted at both time points.

At 12 weeks (Figure S15), the cumulative Z-curve initially crossed the conventional threshold for statistical significance, indicating favorable results. However, it did not cross the O'Brien-Fleming boundary, and the accrued sample size ($n = 443$) remained below the RIS ($n = 468$). These findings indicate a potential short-term benefit but are not conclusive.

TSA revealed a more conclusive pattern at 24 weeks (Figure S16). The cumulative Z-curve crossed the conventional and O'Brien-Fleming thresholds prior to reaching the RIS ($n = 368$), and the final Z-score remained consistently above the efficacy boundary. These results provide statistically robust evidence supporting medium-term functional improvement associated with ablative neurolysis. A summary of the TSA findings is presented in Table 2, detailing which outcomes crossed the monitoring boundaries and their corresponding interpretations.

Bipolar RFA and CRFA showed relatively reduced efficacy among the techniques in most outcomes. A ranking of interventions based on MDs is presented in Table 3. Importantly, all TSA analyses accounted for procedural variables: lesion number per nerve was uniform across studies (1 lesion per target), 7 trials used the classical 3-nerve approach while 1 trial targeted only 2 nerves, and electrode sizes varied (17G, 18G, 22G, 27G). Subgroup analyses stratified by electrode size revealed no statistically significant differences in pain, function, or GPES outcomes, although studies employing 22G electrodes contributed to higher heterogeneity. Therefore, the observed differences across techniques persisted even after considering these procedural parameters.

Discussion

This systematic review and meta-analysis provide robust evidence that genicular neurolysis techniques offer clinically significant pain relief and functional improvement in patients with knee OA compared to sham procedures.[22] These benefits were most pronounced at 12 and 24 weeks, with the pooled effects surpassing the established thresholds for MCID. While bipolar RFA and CRFA appeared less effective than conventional monopolar RFA in our subgroup analyses, these differences persisted even after accounting for electrode size, number of nerves, and number of lesions. Our results, therefore, suggest that procedural parameters are unlikely to explain the observed differences in efficacy across techniques. However, the limited number of trials per modality and the high degree of heterogeneity underscore the need for further head-to-head RCTs with standardized protocols before definitive conclusions can be drawn regarding comparative effectiveness.

The included trials differed in whether a diagnostic genicular nerve block was performed before randomization. While some protocols required a positive block response (eg, ≥50% temporary pain relief) as an eligibility criterion, others proceeded directly to ablation without prior block testing. This variability was not a focus of our primary analysis but may represent a relevant source of heterogeneity in treatment effect, as nerve blocks can identify likely responders and potentially exclude patients with central or extra-articular pain sources. However, the predictive value of diagnostic genicular nerve blocks for long-term outcomes after neurolysis remains uncertain, with limited high-quality evidence and no consensus on threshold response or optimal methodology.[23] Future RCTs should address the role of diagnostic blocks in refining patient selection for genicular neurolysis.

Because electrode size, number of nerves targeted, and lesions per nerve were largely consistent across the included

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trials, these factors are unlikely to explain the observed differences in efficacy among techniques. Nevertheless, the small number of studies per modality, heterogeneity in study design, and absence of direct head-to-head comparisons limit any firm conclusions. Future randomized trials should therefore standardize reporting of procedural parameters to ensure that differences in outcomes reflect the neurolysis modality rather than variation in technical implementation.

Technical aspects of radiofrequency procedures, such as electrode size, number of nerves targeted, and number of lesions applied per nerve, may potentially influence clinical outcomes. In this review, we observed that almost all included trials adopted the conventional approach of targeting 3 genicular nerves, with 3 lesions per nerve, ensuring relative uniformity in these parameters. Electrode size, however, varied across studies (17G, 18G, 22G, and 27G). Subgroup analyses stratified by electrode gauge did not demonstrate statistically significant differences in treatment effect, although trials using 22G electrodes contributed to greater heterogeneity and yielded more extreme estimates. These findings indicate that variability in electrode size may partially account for inconsistencies between trials, but current evidence does not suggest a systematic effect on efficacy. Standardization and transparent reporting of procedural details in future RCTs will be important to improve comparability and to clarify whether technical parameters influence treatment outcomes.

Applying the GRADE framework, we found moderate certainty evidence that genicular neurolysis provides clinically meaningful pain relief at 12 weeks and functional improvement at 24 weeks, with downgrades primarily due to inconsistency across trials. Evidence for long-term outcomes beyond 24 weeks, and for patient satisfaction measures (GPES), was rated as low certainty due to imprecision and heterogeneity. These ratings underscore the need for larger, more standardized RCTs to strengthen the evidence base.

The superiority of conventional monopolar RFA warrants careful consideration. While all evaluated techniques—conventional monopolar RFA, CRFA, bipolar RFA, and cryoneurolysis—aim to disrupt nociceptive transmission from genicular nerves, they differ fundamentally in their biophysical mechanisms.[24] All RFA modalities, including conventional monopolar, cooled, and bipolar techniques, achieve neurolysis through thermal coagulation, differing primarily in lesion geometry rather than mechanism. In contrast, cryoneurolysis induces nerve disruption via ice crystal formation and Wallerian-like degeneration, representing a mechanistically distinct approach. For our analysis, the unifying principle is the creation of neurolysis, regardless of whether it is achieved by heat or cold.[25] In our sham-controlled synthesis, conventional monopolar RFA appeared to demonstrate more consistent therapeutic benefits compared with other modalities. However, these findings should be interpreted cautiously, as differences in study design, guidance methods, and procedural techniques may also have influenced outcomes. Given the limited number of trials and absence of direct head-to-head comparisons, no firm conclusion can be drawn regarding the superiority of one technology over another. Future randomized and comparative trials, or network meta-analyses, will be necessary to better determine whether specific RFA modalities offer distinct clinical advantages.

Our findings suggest that the precision of cannula placement at anatomically validated landmarks may play an equally or even more important role than lesion size in determining clinical outcomes. While most included trials reported image-guided targeting of the superomedial, superolateral, and inferomedial genicular nerves, some lacked sufficient detail on the specific anatomical landmarks applied. One study (Radnovich et al[18]) used anatomical landmarks without imaging guidance, highlighting the variability in protocols. Such heterogeneity underscores the need for standardized and validated targeting methods, as inaccurate placement may negate the potential benefits of larger lesioning technologies. Future research should therefore prioritize anatomically rigorous approaches when evaluating and comparing RFA modalities, to ensure that differences in efficacy reflect the technology rather than variation in targeting accuracy.

A key strength of this analysis was the use of TSA to assess the robustness of the evidence while minimizing the risk of random error. TSA functions as a statistical safeguard in cumulative meta-analyses, allowing conclusions to be drawn with greater confidence when the cumulative Z-curves cross predefined monitoring boundaries before the required information size is reached.[13] In our study, the evidence for pain reduction at 12 weeks and functional improvement at 24 weeks crossed these boundaries, indicating that the short- to medium-term benefits are robust and unlikely to be overturned by future trials of similar design. In contrast, for long-term pain relief beyond 24 weeks and functional improvement at 12 weeks, although conventional meta-analysis showed statistical significance, TSA neither met the required information size nor crossed the efficacy thresholds. These outcomes should therefore be interpreted as promising but inconclusive, underscoring the need for additional high-quality RCTs with extended follow-up to confirm the durability of effects.

Significant inter-study heterogeneity was a major finding, with I^2 values frequently exceeding 80%. This variability likely stems from methodological inconsistencies across trials, including differences in patient selection (eg, Kellgren-Lawrence grades), procedural protocols such as nerve-targeting techniques (anatomic vs image-guided), and lesioning parameters. While nearly all studies employed image guidance with either fluoroscopy or ultrasound, Radnovich et al[18] relied solely on anatomical landmarks. Although this isolated methodological difference could not be tested through subgroup analyses, it highlights how variations in targeting approach may contribute to heterogeneity and underscores the need for standardized, image-guided protocols and consistent reporting in future clinical trials of genicular nerve ablation.

Our findings regarding patient satisfaction, measured using the GPES, were inconclusive and did not show a significant benefit. This may be attributable to the limited number of studies reporting this outcome, the use of inconsistent measurement tools, and the powerful influence of nonspecific effects, such as patient expectations and procedural context, on subjective reports.

This study had some limitations. First, the external validity of our findings may be constrained, as most included trials recruited patients with chronic refractory OA from tertiary care centers, which may not represent patients in the earlier stages of the disease.[26] Second, long-term efficacy and safety data beyond 1 year are scarce. Third, no trials have incorporated cost-effectiveness analyses or biomarker-informed

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strategies to identify patient phenotypes that are most likely to respond to a specific technique. Finally, while the overall safety profile was favorable, with no serious adverse events reported, the potential for underreporting existed due to inconsistent monitoring protocols.

Conventional RFA showed the most consistent benefit in sham-controlled comparisons; however, this finding should not be interpreted as technological superiority over cooled RFA, bipolar RFA, or cryoneurolysis. Our subgroup analyses rely on indirect comparisons and are therefore limited in their ability to establish comparative effectiveness. Standardized head-to-head trials and network meta-analyses will be necessary to directly address differences between these modalities.

Conclusions

This meta-analysis confirmed that genicular nerve ablation procedures are effective for managing pain and disability in knee OA, with consistent evidence of pain relief at 12 weeks and functional improvement at 24 weeks. Among sham-controlled trials, conventional monopolar RFA demonstrated the most robust and reproducible benefits. However, significant evidence gaps remain regarding long-term durability, comparative effectiveness across technologies, and optimal patient selection. Importantly, because direct head-to-head comparative trials are lacking, our findings cannot establish the relative superiority of one RFA modality over another.

Future research should prioritize well-designed, adequately powered RCTs—including active comparator and head-to-head designs—with standardized protocols and outcome measures. Such efforts will be essential to refine the role of conventional, cooled, bipolar RFA, and cryoneurolysis in personalized OA care and ensure that technical and procedural factors are consistently reported.

Supplementary material

Supplementary material is available at *Pain Medicine* online.

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